Venkata Lakshmi Parimala Pasupuleti

Senior Software Engineer · Affordable Housing & Multifamily PropTech · HUD Compliance · LIHTC / Tax Credit · Section 8 / HAP · Python · Java · PHP · PostgreSQL · AWS

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PROFESSIONAL SUMMARY

Senior Software Engineer with **8 years of specialized expertise in affordable housing and multifamily PropTech** at Entrata — one of the leading property management SaaS platforms — owning the systems that sit at the intersection of software engineering and housing compliance. Built and maintained the **HUD annual income-limit import pipeline** for nearly 7 years: a few thousand federal source rows expanded into ~180,000 domain records powering **Section 8 / HAP eligibility, LIHTC rent calculations, tax-credit certifications, and Area Median Income (AMI) thresholds** for 100K+ housing units across 7 consecutive regulatory cycles. Owned the **RabbitMQ-based async gross-rent update system** processing annual rent-schedule changes across thousands of tenant records. Led the **affordable housing module integration for the AWS S3 document migration** — resident certification documents, TIC forms, HAP contracts, and compliance records — with 100% PHI compliance and zero security incidents. Deep understanding of **Tenant Income Certification (TIC), annual and interim recertifications, move-in/move-out processing, utility allowances, wait list management, HUD 50059, and regulatory reporting** workflows — not as a feature engineer, but as the long-term system owner through seven years of production incidents and annual regulatory cycles. Complemented with recent AI work: built **Leasel**, an AI platform for lease and compliance document intelligence. Python, Java, PHP, JavaScript/TypeScript, PostgreSQL, AWS, Docker, CI/CD. M.S. Data Analytics Engineering, George Mason University (GPA 3.87).

CORE COMPETENCIES

Affordable Housing & Multifamily PropTech | HUD Compliance & Income Limits (AMI) | LIHTC / Tax Credit (Section 42) | Section 8 / HAP / Housing Assistance Payments | Tenant Income Certification (TIC) | Annual & Interim Recertifications | HUD 50059 & Compliance Reporting | Rent Calculations & Utility Allowances | Wait List Management | Resident Document Management | Multi-Tenant Property Management SaaS | Python / FastAPI / Django | Java / Spring Boot | PHP / Laravel | JavaScript / TypeScript / React | PostgreSQL & SQL Optimization | AWS (S3 / Lambda / RDS / SQS) | Async Processing (RabbitMQ) | REST APIs & Microservices | CI/CD & Docker / Kubernetes | PHI / HIPAA Compliance | AI-Enabled Compliance Features (RAG / LLMs)

EXPERIENCE

Blue Horse Digital

Sep 2026 - Present

AI/ML Engineer (Volunteer) — Virginia, USA

- Building AI-enabled applications and data pipelines; integrating AI components with backend and frontend systems; surfacing business insights through application-facing APIs.

Magizhs Tech Solution

May 2025 - Aug 2025

AI/ML Engineer Intern

- Built an agentic workflow system (LangGraph stateful graph) with RAG over a regulatory corpus, validation guardrails, and MCP server deployment for downstream tool integration.

Entrata — PropTech / Real Estate SaaS

Sep 2018 - Aug 2024

Senior Software Engineer — Lehi, UT (Remote)

- **Long-term owner of the HUD annual income-limit import pipeline** — transformed federal HUD/USDA income-limit tables into ~180,000 domain records powering **Section 8 HAP eligibility, LIHTC rent calculations (Section 42), tax-credit certifications, and Area Median Income (AMI) thresholds** for 100K+ housing units across 7 consecutive regulatory cycles. Every release included baseline validation, reconciliation between current and prior-year limits, and rollback preparation - a wrong import means thousands of residents receive incorrect eligibility decisions or rent calculations. Zero compliance violations across the full ownership period.
- **Built and owned compliance workflow modules** covering **Tenant Income Certification (TIC), annual and interim recertifications, move-in/move-out processing, utility allowance calculations, and wait list management** — the full lifecycle of a resident in an affordable housing program. Designed data models and business logic to handle the rule differences between Section 8, LIHTC, HOME, and Rural Development programs within a single multi-tenant platform.
- **Owned the RabbitMQ-based async gross-rent update system** propagating annual rent-schedule changes (including utility allowance adjustments) across thousands of tenant records — designed retry logic and idempotency guarantees eliminating double-processing failures; maintained through production incidents and annual HUD/LIHTC rent-schedule cycles.
- **Led the Affordable Housing module integration for the AWS S3 document migration** — resident certification documents, TIC forms, HAP contracts, compliance reports, and housing authority correspondence; 100% PHI compliance with Libsodium-encrypted pipelines; zero security incidents across the full resident data migration.
- **Shipped 20+ full-stack production modules** (Python, PHP, Java, PostgreSQL, JavaScript, RabbitMQ) across 120+ Agile sprints; led microservices redesign (Docker) — 50% scalability improvement, 30% deployment time reduction; eliminated 80% of manual operational workload through automation pipelines.
- Optimized 200+ SQL queries (30% performance gain) across eligibility, rent calculation, certification, and document management schemas; mentored 5+ engineers; conducted 300+ code reviews.

Entrata India — Db Xento Systems

Jan 2017 - Sep 2018

Associate / Trainee Software Engineer — Pune, India

- Built affordable housing compliance modules (PHP, Python, Java, PostgreSQL) — **HUD/Tax Credit income-limits import (1.6M records across 5 years)**, resident document management, TIC and certification workflows, and state-specific housing certification documents for all 56 US states/territories.
- Led cloud migration of resident compliance documents and HAP contract records from legacy server storage to AWS S3 — resolved 90% of legacy space constraints; redesigned document storage schema migrating hundreds of thousands of multifamily housing records.

PROJECTS

Leasel: AI-Powered Affordable Housing Lease Compliance & Document Intelligence

HUD · LIHTC · RAG · Compliance · LangGraph · Document Processing

Production AI platform built for the **affordable housing compliance** use case — the same document and regulatory complexity found in multifamily HUD/LIHTC platforms: OCR-fallback ingestion, section-aware chunking, **hybrid RAG** (TF-IDF + semantic reranking), **citation grounding** to exact lease and regulatory clauses, missing/conflicting clause detection. **8-agent LangGraph workflow** (Triage → Retrieval → Configuration → Historical Incident → Structured Data → Reasoning → Validation → Human Escalation) with conditional HITL routing when AI confidence is insufficient. Evaluated on **residential lease PDFs across 5 US states + federal HUD/Section 8 regulatory documents**. Core rule enforced in code: LLM extracts and reasons; a human makes every eligibility or compliance decision. Eval: **93.6% precision, 87.9% F1** on 1,240 test cases; CI regression gate.

Python · LangGraph · FastAPI · TF-IDF · ChromaDB · OCR · SQLite · GitHub Actions · 2026

AgentProbe: Open-Source AI Evaluation Toolkit

LLM Evaluation · Multi-Provider · CI/CD · Hallucination Detection

Reusable toolkit for evaluating AI features in production: **5-metric eval engine** (correctness, groundedness, latency, token efficiency, hallucination rate), LLM-as-judge scoring, CI-integrated regression harness with golden datasets, offline + online evaluation loops, multi-provider routing (OpenAI, Anthropic, Groq, Ollama). Demonstrates the engineering discipline applied to AI-enabled product features — ensuring they are accurate, reliable, and observable before and after deployment. Live: github.com/venkatalakshmiparimala/AgentProbe

Python · FastAPI · LangGraph · ChromaDB · CI/CD (GitHub Actions) · 2025

CrashClarity: Large-Scale Road Accident Data Analysis (104K Records)

ML Pipeline · Python · scikit-learn · Data Analysis

End-to-end ML analysis pipeline on 104K+ UK road accident records — feature engineering across weather, road condition, and time variables; Random Forest + Logistic Regression model (85%+ accuracy); interactive dashboard deployed on GitHub Pages. Demonstrates data engineering and statistical modeling applied to large real-world datasets.

Python · R · scikit-learn · Random Forest · Logistic Regression · GitHub Pages · 2025

EDUCATION

M.S. Data Analytics Engineering — George Mason University, **Virginia** — May 2026 | GPA 3.87

B.Tech Computer Science & Engineering — JNTU, **Anantapur** — 2017